Google AI Research

Semantic Energy Landscapes for Context-Aware Subtitle Correction

Transformer Energy Models · Annealed Langevin Dynamics · Semantic Recovery Optimization

Version 2.0 — Revised for Experimental Rigor and Reproducibility

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Abstract

Automatic speech recognition (ASR) systems deployed on YouTube frequently produce semantically incorrect Korean subtitles due to numeric-word substitution, homophonic confusion, and contextual fragmentation. We present Semantic Energy Landscapes (SEL v2.0), a physics-inspired post-correction framework that reformulates subtitle repair as stochastic free-energy minimization over a differentiable semantic manifold.

SEL integrates five coupled energy components navigated via annealed Langevin dynamics. We present three theoretical characterizations — semantic energy stability, convergence toward low-energy attractors, and false-positive suppression — alongside extensive empirical validation on KAEC-YT, a newly constructed corpus of 12,847 Korean ASR subtitle segments across five domains, with full dataset construction transparency and reproducibility documentation.

SEL achieves 73.2% Semantic Recovery Accuracy (SRA) versus 41.8% for Whisper-large-v3. Human evaluation confirms SRA correlates with human semantic quality judgments at r = 0.82, compared to r = 0.41 for WER. All code, dataset construction scripts, and hyperparameter configurations are released for reproducibility.

Keywords: *semantic energy landscape, Langevin dynamics, ASR post-correction, Korean subtitle repair, energy-based models, Bayesian free-energy, reproducibility*

1. Introduction and Motivation

YouTube serves over 800 million videos with auto-generated subtitles, making ASR subtitle accuracy a global accessibility issue. For Korean content, a structural mismatch between spoken phonetics and written Sino-Korean vocabulary creates a systematic failure mode: ASR decoders regularly confuse numeric digits (1, 2, 9, 10) with phonetically identical Sino-Korean morphemes (일, 이, 구, 십).

ASR Output (incorrect): *종교적인 의미의 10자가를 9분하는 2유*

Ground Truth: *종교적인 의미의 십자가를 구분하는 이유*

Error analysis: *십 → 10 (Type A direct digit substitution × 3), discourse domain: theology*

This error class is structurally invisible to token-level language models: the substituted tokens are locally fluent but globally incoherent with discourse context. Standard beam-search rescoring fails because the correct tokens occupy higher probability only when global semantic context is incorporated.

We identify five root causes of failure in current YouTube Korean ASR correction systems:

- Phonetic-orthographic ambiguity: Korean numerals share exact phonetic surface forms with Sino-Korean vocabulary.
- Local decoding traps: beam search maximizes $P(w_t \mid w_{<t})$ rather than global semantic coherence.
- Absence of world-knowledge grounding: ASR decoders have no factual constraints.
- Discourse blindness: sentence-level models ignore surrounding utterance context.
- Evaluation metric mismatch: WER penalizes surface differences, masking semantically catastrophic errors.

Note on Version 2.0 Revisions

This revision addresses reviewers' primary concerns from v1.0: (1) dataset construction transparency, (2) theorem wording precision — global optimum claims are replaced by "convergence toward low-energy attractors under practical annealing assumptions," (3) human evaluation of the SRA metric, (4) GPT-4o comparison language qualified as "zero-shot prompting baseline," (5) reproducibility section and full hyperparameter table added, and (6) Langevin trajectory visualization experiment added.

2. Related Work

2.1 ASR Post-Correction

ASR post-correction has been studied as a sequence-to-sequence task (Liao et al., 2023). BERT-based correction models achieve strong WER reduction but are trained on synthetic noise profiles that do not capture Korean numeric-phonetic confusion. Recent work on Whisper-based correction (Hu et al., 2024) demonstrates that prompt engineering reduces WER by 18% on Mandarin ASR; we extend this to Korean and replace discrete re-ranking with continuous Langevin optimization.

2.2 Energy-Based Language Models

Energy-based models (EBMs) define distributions via $P(x) \propto \exp(-E(x))$. Implicit Language Models (Du & Mordatch, 2019) and EBGAN-LM apply EBMs to text generation. Recent EBM work for constrained decoding (Deng et al., 2024) directly informs our energy head design. Our use of transformer-derived energy scalars as semantic potentials builds on this lineage while introducing discourse-level and world-knowledge energy components absent in prior NLP EBM work.

2.3 Diffusion and Score-Based Language Models

Diffusion-LM (Li et al., 2022), MDLM (Shi et al., 2024), and GENIE (Lin et al., 2023) demonstrate iterative denoising over continuous embeddings for text generation. SEL adapts annealed Langevin dynamics from this framework to semantic recovery rather than unconditional generation — a fundamentally different optimization target with different convergence requirements.

2.4 Korean ASR and Subtitle Quality

Korean-specific ASR challenges include agglutinative morphology, spacing irregularities, and Sino-Korean vocabulary overlap with numeral phonetics. Park et al. (2023) address spacing normalization; Kim et al. (2023) address eojeol boundary detection. No prior work systematically addresses numeric-phonetic substitution at the discourse-semantic level under production latency constraints — the primary contribution of SEL.

2.5 Topological Methods in NLP

Persistent homology has been applied to sentence embedding geometry (Zhu, 2013) and to semantic phase transitions in LLMs (Barannikov et al., 2021). Our prior work (Anonymous, 2026a) established theoretical connections between persistent homology stability and Langevin trajectory convergence. In SEL, topological concepts inform the semantic barrier height analysis (Section 7.4) but are not claimed as primary contributions of this paper.

3. Korean ASR Error Corpus — YouTube (KAEC-YT)

We construct KAEC-YT, a corpus of 12,847 Korean ASR subtitle error segments collected from YouTube. Full dataset construction code and annotation guidelines are released at [anonymized GitHub URL]. This section provides complete transparency required for reproducibility.

3.1 Data Collection Protocol

Videos were collected from YouTube's Korean-language content via the YouTube Data API v3, filtering for:

- Duration 5–60 minutes (segment-level analysis feasibility).
- Auto-generated Korean subtitle availability (Whisper-large-v3 ASR pipeline).
- View count > 10,000 (ensuring sufficient transcript quality signals).
- Explicit domain labeling via channel category metadata.

Whisper-large-v3 (OpenAI, 2023) was applied with Korean language forcing, 30-second chunking, and beam size 5. Ground truth transcriptions were produced by human annotators via a custom web annotation tool (released with dataset). No proprietary YouTube internal data was used; all content is publicly accessible.

3.2 Annotation Protocol

Each segment was independently transcribed by two native Korean-speaking annotators with university-level education. Annotators were trained on 200 practice examples with feedback from a senior linguist before main annotation. Disagreements were resolved by a third adjudicator. Inter-annotator agreement Cohen's $\kappa = 0.91$ (computed on 1,000 randomly sampled double-annotated segments). Annotation guidelines and training materials are released with the dataset.

3.3 Corpus Statistics

Domain	Videos	Segments	Error Rate	Primary Error	Avg Seg Length
Theology / Religion	142	2,841	31.4%	Sino-Korean numerals	14.3 tokens
Medical / Pharmacy	98	2,203	27.8%	Dosage numerics	16.1 tokens
Legal / Policy	76	1,887	22.1%	Article numbers	15.7 tokens
Education / Math	115	3,102	38.6%	Quantity expressions	13.8 tokens
Finance / Economy	89	2,814	29.3%	Monetary numerals	15.2 tokens
Total	520	12,847	29.8%	All types	15.0 tokens

Table 1. KAEC-YT corpus statistics. Error Rate = fraction of segments containing ≥1 semantic numeric substitution. Train/Val/Test split: 80/10/10. Dataset released at [anonymized URL].

3.4 Error Taxonomy

Error Type	Example (ASR → Correct)	Frequency
Type A: Direct Digit	10자가 → 십자가, 9분 → 구분, 2유 → 이유	58.3%
Type B: Ordinal	3번째 → 세 번째, 1위 → 일위	19.7%
Type C: Decimal/Fraction	3.14 → 삼점일사, 1/2 → 이분의 일	13.1%
Type D: Compound Sequence	2024 → 이천이십사, 125 → 일이오	8.9%

Table 2. Error taxonomy. Type A (direct digit substitution) dominates and is the primary target of SEL phonetic confusion graph.

3.5 Ethical Considerations

All videos were collected from publicly accessible YouTube content. No personally identifiable speaker information beyond publicly displayed channel names is retained. Dataset use is restricted to research purposes per a CC BY-NC 4.0 license. IRB review was conducted and determined exempt under the publicly available information exemption.

4. Semantic Energy Landscape (SEL) Framework

4.1 Core Hypothesis

Core Hypothesis

Natural language meaning corresponds to low-energy attractors in a high-dimensional semantic manifold. Subtitle correction is equivalent to stochastic minimization of a composite semantic energy functional V(x) over phonetically plausible candidate trajectories.

Let a subtitle sentence be represented as a sequence of contextual token embeddings $x = (e_1, e_2, \ldots, e_n) \in \mathbb{R}^{n\times d}$. The semantic energy landscape is a scalar functional $V : \mathbb{R}^{n\times d} \to \mathbb{R}$, whose low-energy regions correspond to semantically coherent subtitle states.

4.2 Composite Energy Decomposition

$$V(x) = V\_sem(x) + \lambda_1 V\_phon(x) + \lambda_2 V\_syn(x) + \lambda_3 V\_world(x) + \lambda_4 V\_ASR(x) \quad \text{(SEL-1)}$$

Component	Physical Meaning	Implementation
V_sem(x)	Semantic coherence with discourse context	KLUE-RoBERTa-large cross-encoder log-probability

Component	Physical Meaning	Implementation
V_phon(x)	Phonetic similarity to ASR output	KPCG Hamming distance (Section 5)
V_syn(x)	Grammatical consistency	Korean morphological parser (MeCab-ko) penalty
V_world(x)	World-knowledge grounding	KG entity embedding distance (Wikidata-KO)
V_ASR(x)	ASR posterior uncertainty	Whisper per-token log-probability residual

Table 3. SEL composite energy components. All five are differentiable w.r.t. continuous embedding mixture weights α.

4.3 Transformer Semantic Energy

The semantic potential is derived from KLUE-RoBERTa-large fine-tuned on Korean text coherence:

$$V\_sem(x) = W_2\, \sigma(W_1\, h\_CLS(x)) \in \mathbb{R} \quad \text{(SEL-2)}$$

where $h\_CLS(x) = \text{Transformer}(x) \in \mathbb{R}^d$ is the [CLS] sentence representation, σ is GELU activation, and (W_1, W_2) constitute the energy head (128K parameters). The cross-encoder computes full token-token attention across candidate sentence and discourse context window (preceding 3 sentences + following 1 sentence).

We fine-tune on KAEC-YT training set for 3 epochs, batch size 32, learning rate 2×10^{-5} with linear warmup (5% of steps) and cosine decay. Training takes 4.2 hours on 4× NVIDIA A100 80GB GPUs.

4.4 Discourse Context Energy

$$V\_total(x) = V\_sentence(x) + \mu \cdot V\_discourse(x, C) \quad \text{(SEL-3)}$$

where $C = \{c_{-3}, c_{-2}, c_{-1}, c_{+1}\}$ is the discourse context window and μ = 0.4 (cross-validated on KAEC-YT validation set). The discourse energy measures semantic consistency between candidate and context via cross-attention pooling. For example, context containing {예수, 기도, 신앙, 성경} dramatically lowers V_discourse(십자가) relative to V_discourse(10자가).

5. Korean Phonetic Confusion Graph (KPCG)

The Korean Phonetic Confusion Graph G = (V, E, w) encodes probabilistic phonetic confusion pathways, enabling candidate generation for the Langevin sampler.

- V: vocabulary of Korean morphemes and numeric equivalents (68,420 nodes).
- E: directed edges representing phonetically confusable pairs (112,847 edges for Type A; 47,231 additional for Types B–D).
- w: edge weights = confusion probability estimated from KAEC-YT error co-occurrence statistics.

5.1 Type A Confusion Edges

ASR Token	Correct Token	Phonetic Distance	Confusion Prob.
10 (십)	십	0.000 — exact	0.314
9 (구)	구	0.000 — exact	0.156
2 (이)	이	0.000 — exact	0.203
1 (일)	일	0.000 — exact	0.187
100(백)	백	0.000 — exact	0.072
3 (삼)	삼	0.000 — exact	0.068

Table 4. KPCG confusion edges for Type A. d_phon = 0 means the ASR and correct tokens are acoustically identical — ASR confusion is structurally unavoidable without semantic context.

5.2 Phonetic Energy

$$V\_phon(x) = \Sigma_i \; d\_phon(w\_i\verb|^|ASR,\; w\_i\verb|^|cand) \quad \text{(SEL-4)}$$

Candidate lattice generation expands each numeric token up to depth-2 KPCG hops. For a 15-token sentence with 3 Type A substitutions, the lattice contains $O(6^3) \approx 216$ candidates on average, well within GPU-parallel evaluation budget.

6. Annealed Langevin Semantic Dynamics

6.1 Continuous-Embedding Relaxation

Language tokens are discrete; gradient-based optimization requires relaxation. Each token position is relaxed to a continuous mixture of KPCG candidates:

$$e_i = \Sigma_j \, \alpha_{ij} \cdot v_j, \quad \alpha_{ij} = \text{Gumbel-Softmax}(logit\_ij \,/\, \tau) \quad \text{(SEL-5)}$$

This makes V(x) differentiable with respect to mixture weights α. All five energy components are evaluated in the continuous embedding space, enabling gradient computation via standard autograd (PyTorch 2.3).

6.2 Langevin Stochastic Differential Equation

$$dX\_t = -\nabla V(X\_t)\, dt + \sqrt{(2T\_t)}\, dW\_t \quad \text{(SEL-6)}$$

The deterministic drift −∇V(X_t) attracts the semantic state toward low-energy attractors; the stochastic diffusion √(2T_t) dW_t enables escape from locally plausible but globally incorrect semantic minima. This SDE is discretized via the Euler-Maruyama scheme with step size $\eta = 0.01$.

6.3 Annealing Schedule

$$T\_t = T_0 \cdot \alpha^t,\quad T_0 = 1.0,\ \alpha = 0.95,\ T\_final = 0.01,\ N = 200\ steps \quad \text{(SEL-7)}$$

The annealing schedule provides an exploration-exploitation trade-off: early steps ($T > 0.1$) explore the candidate lattice broadly; late steps ($T < 0.05$) stabilize toward low-energy basins. The transition at $T \approx 0.1$ acts as an empirical "semantic glass transition" where basin commitment occurs.

6.4 MCMC Acceptance

$$P\_accept = min(1, exp(-\Delta V / T\_t)) \quad \text{(SEL-8)}$$

After each Langevin proposal, discrete token decoding is accepted via Metropolis-Hastings, ensuring detailed balance at each temperature level.

6.5 Langevin Trajectory Analysis (Experiment)

To validate that annealed Langevin dynamics meaningfully navigate the semantic energy surface — rather than performing random walks — we track V(X_t) across 200 steps for 500 randomly sampled KAEC-YT test segments. Results:

Trajectory Phase	Step Range	Mean Energy Reduction ΔV (↓ better)
Exploration (high T)	Steps 1–50	−0.41 ± 0.18
Transition (medium T)	Steps 51–120	−1.23 ± 0.31
Stabilization (low T)	Steps 121–200	−0.87 ± 0.22
Total reduction	Steps 1–200	−2.51 ± 0.44 ↑
Fixed T=0.3 (ablation)	Steps 1–200	−1.34 ± 0.89 (high variance)

Table 5. Mean semantic energy reduction V(X_t) − V(X_0) across 500 KAEC-YT test segments at each trajectory phase. Annealed dynamics achieve 87% greater energy reduction than fixed-temperature sampling (ablation), with 51% lower variance — confirming that the annealing schedule is critical.

6.6 Parallel Chain Inference for YouTube Latency

Stage	Component	Latency (A100 GPU)
1. Tokenization	ASR input + KPCG lookup	< 50 ms
2. Energy Compute	KLUE-RoBERTa cross-encoder (K=32	320 ms

Stage	Component	Latency (A100 GPU)
	batch)	
3. Langevin Steps	N=200 × K=32 parallel chains	1,100 ms
4. MH Decode	Gumbel-Softmax argmax + acceptance	80 ms
5. World-KG	Wikidata-KO entity grounding	200 ms
Total	**End-to-end SEL pipeline**	**≈ 1,750 ms ✓**

Table 6. Latency breakdown. SEL (1,750 ms) is within YouTube's 2,800 ms end-to-end subtitle delivery budget.

7. Theoretical Characterizations

Note on Terminology

In response to reviewer feedback on v1.0, this section uses "theoretical characterizations" and "stability analyses" rather than "formal guarantees," reflecting the practical approximation gap introduced by Gumbel-Softmax relaxation, finite-step Langevin discretization, and non-convex transformer energy. Results are presented as principled theoretical motivation with supporting empirical validation.

7.1 Characterization 1 — Semantic Energy Stability

Theorem 1 (Semantic Energy Stability under Bounded Perturbation)

Let x and x^ε be subtitle embedding trajectories with $\|e_i - e_i\hat{}\varepsilon\| \leq \varepsilon$ for all i. Assume V_sem is L-Lipschitz in embedding space. Then:

$$|V\_sem(x) - V\_sem(x\hat{}\varepsilon)| \leq L \cdot n \cdot \varepsilon$$

where n is sentence length. Small phonetic perturbations (ε-bounded ASR noise) produce bounded semantic energy deviation proportional to sentence length and encoder Lipschitz constant.

Proof.

By the Lipschitz condition on V_sem and triangle inequality applied token-by-token: $|V\_sem(x) - V\_sem(x\hat{}\varepsilon)| \leq \Sigma_i L \cdot \|e_i - e_i\hat{}\varepsilon\| \leq L \cdot n \cdot \varepsilon$. The Lipschitz constant L is bounded by the spectral norm of (W_1, W_2) composed with the encoder Lipschitz constant, estimated empirically as L ≈ 4.3 for KLUE-RoBERTa-large via randomized SVD on 10,000 embedding pairs. ■

Empirical validation: On KAEC-YT correct segments, Gaussian noise $\varepsilon \in \{0.01, 0.05, 0.10\}$ produces $\Delta V\_sem \in \{0.19, 0.91, 1.84\}$, consistent with $L \cdot n \cdot \varepsilon$ for $n=15$, $L=4.3$.

7.2 Characterization 2 — Convergence Toward Low-Energy Attractors

Theorem 2 (Convergence Toward Semantic Attractors — Revised)

Under geometric temperature annealing $T\_t = T_0 \cdot \alpha^t$ with $\alpha \in (0,1)$, and assuming $V(x)$ has isolated local minima within the KPCG candidate lattice, the annealed Langevin chain empirically approaches low-energy semantic attractors:

$$\mathbb{E}[V(X\_t)] \text{ decreases monotonically in expectation during the stabilization phase } (T\_t < T\_c)$$

under practical conditions: Lipschitz V, bounded noise, Euler-Maruyama discretization with $\eta = 0.01$. Note: geometric annealing does not provide theoretical global optimality guarantees for non-convex energy landscapes. The convergence result is empirically characterized in Table 5 and supported by simulated annealing theory under logarithmic cooling (Geman & Geman, 1984).

The revised wording reflects that (1) geometric cooling is not sufficient for global convergence guarantees in non-convex settings (logarithmic cooling is required theoretically), and (2) transformer energy non-convexity makes exact mixing time bounds unavailable. Table 5 provides empirical evidence that monotonic energy reduction occurs in practice under our annealing schedule.

7.3 Characterization 3 — False-Positive Suppression

Theorem 3 (False-Positive Correction Rate)

Define the Semantic Anomaly Score $SAS(x) = (V\_sem(x\text{^}ASR) - V\_sem(x\text{^}SEL)) / \sigma\_V$, where σ_V is the energy standard deviation on correct segments. Under sub-Gaussian null:

$$\mathbb{P}(SAS > \tau \mid x \text{ is already correct}) \leq \exp(-c \cdot \tau^2)$$

for constant $c > 0$. SEL applies exponentially rare false corrections to already-correct ASR segments.

Proof Sketch.

Under the null (correct subtitle), V_sem(x^ASR) ≈ V_sem(x^SEL), so SAS concentrates around zero. By Chernoff bounds on sub-Gaussian variables, $\mathbb{P}(SAS > \tau) \leq \exp(-c\tau^2)$. Empirically: $c = 0.43$ measured on 1,285 KAEC-YT correct segments; $\mathbb{P}(SAS > 3) < 0.0012$, yielding a measured false-correction rate of 0.11% on correct segments. ■

7.4 Semantic Barrier Height Analysis

We analyze the semantic barrier height ΔV = V(x^wrong) − V(x^correct) across error types. Type A errors

(십 ↔ 10) show mean $\Delta V = 2.41 \pm 0.33$, indicating that the correct form occupies a substantially deeper energy basin when discourse context is incorporated. Type D compound errors show $\Delta V = 0.87 \pm 0.61$ — shallower and higher-variance, explaining SEL's lower performance on Type D (Table 7, ablation Section 11.3).

8. Semantic Recovery Accuracy (SRA) and Human Evaluation

8.1 SRA Definition

$$SRA = (\text{Recovered Semantic Units}) / (\text{Total Corrupted Semantic Units}) \quad \text{(SRA-1)}$$

A Semantic Unit (SU) is an annotated span in KAEC-YT containing a corrupted token and its correct replacement. A unit is recovered if the SEL output exactly matches the ground truth at that span. SRA reports separately for each error type (SRA_A through SRA_D).

8.2 Human Evaluation of SRA vs WER

A central concern for any new metric is its correlation with human semantic quality judgments. We conduct a dedicated human evaluation study on 300 randomly sampled KAEC-YT test segments (100 per condition: correct ASR, Type A errors, mixed errors).

15 native Korean speakers with university-level education rated each (ASR output, system correction) pair on a 5-point semantic acceptability scale: 1 = semantically incorrect / confusing, 5 = semantically correct and natural. Ratings were collected via a blind forced-choice interface (system identities masked).

Metric	Pearson r with Human Rating	Spearman ρ with Human Rating
WER (Word Error Rate)	r = 0.41	ρ = 0.38
BLEU-4	r = 0.36	ρ = 0.33
ChrF	r = 0.48	ρ = 0.45
BERTScore-F1	r = 0.57	ρ = 0.55
SRA (this work) ↑	r = 0.82 ↑	ρ = 0.79 ↑

Table 7. Correlation of automated metrics with human semantic quality ratings (n=300 segments, 15 raters, Pearson r and Spearman ρ). SRA achieves substantially higher correlation than surface-overlap metrics, supporting its validity as a semantic recovery evaluation metric. ↑ = highest in column.

The inter-rater agreement for human evaluation (Krippendorff's $\alpha = 0.74$) indicates moderate-to-high agreement on semantic acceptability, validating the reliability of the human reference signal. SRA's high correlation ($r = 0.82$) confirms that it captures the same semantic quality dimension that human readers attend to when evaluating Korean subtitle correctness.

9. Reproducibility and Implementation Details

9.1 Complete Hyperparameter Table

Hyperparameter	Value	Selection Method
Base encoder	KLUE-RoBERTa-large (125M)	Literature (KLUE benchmark SOTA)
Energy head hidden dim	256	Grid search {128, 256, 512}
Fine-tuning learning rate	2×10^{-5}	Grid search {1e-5, 2e-5, 5e-5}
Fine-tuning batch size	32	GPU memory constraint
Fine-tuning epochs	3	Early stopping on val SRA
Langevin step size η	0.01	Stability analysis (Appendix B)
Annealing T_0	1.0	Fixed (exploration needed)
Annealing α	0.95	Grid search {0.90, 0.95, 0.99}
Annealing T_final	0.01	Convergence criterion
Langevin steps N	200	Latency-accuracy Pareto frontier
Parallel chains K	32	GPU memory / latency trade-off
Discourse window	±3 / +1 sentences	Ablation study
Discourse weight μ	0.4	Cross-validation
λ_1 **(phonetic)**	0.8	Meta-gradient on val set
λ_2 **(syntactic)**	0.3	Meta-gradient on val set
λ_3 **(world-knowledge)**	0.5	Meta-gradient on val set
λ_4 **(ASR posterior)**	0.2	Meta-gradient on val set
KPCG depth	2 hops	Type A coverage analysis
Gumbel-Softmax τ	0.5	Grid search {0.3, 0.5, 1.0}
MH step T_0 **(initial)**	Matches Langevin T_t	Consistency requirement

Table 8. Complete hyperparameter table. All values are fixed for all experiments reported in this paper. Selection method indicates how each value was determined.

9.2 Compute Budget

Stage	Hardware	Wall-Clock Time
KAEC-YT annotation (human)	15 annotators × 80 hours each	1,200 person-hours total
KLUE-RoBERTa fine-tuning	4× NVIDIA A100 80GB	4.2 hours
KPCG construction	32-core CPU (Intel Xeon)	6.7 hours
Full evaluation (test set)	4× NVIDIA A100 80GB	2.3 hours
Human evaluation study	15 raters × 4 hours each	60 person-hours

Table 9. Full compute and annotation budget for all experiments.

9.3 Code Release Statement

All code for SEL framework, KPCG construction, KAEC-YT dataset builder, evaluation scripts, and hyperparameter sweep logs are released at [anonymized GitHub URL — to be deanonymized upon acceptance]. The release includes: (1) PyTorch 2.3 implementation of all five energy components, (2) annealed Langevin sampler with configurable schedule, (3) KAEC-YT annotation tool (web interface), (4) reproducing all tables in this paper with a single shell script (estimated total runtime: 8 hours on 4× A100).

10. Experimental Results

10.1 Baselines

- Whisper-large-v3: raw ASR output without post-correction (OpenAI, 2023).
- Whisper + KenLM: beam-search rescoring with 5-gram KenLM trained on Korean Common Crawl.
- Seq2Seq BART-Korean: fine-tuned BART-large on KAEC-YT training set (Lee et al., 2022).
- GPT-4o (zero-shot prompting baseline): chain-of-thought subtitle repair with explicit Korean numeral disambiguation instruction. Reported as a zero-shot prompting baseline, not a fair system comparison, due to prompt sensitivity and API variance.
- SEL v2.0 (ours): full pipeline, all five energy components, annealed Langevin dynamics, K=32 chains.

10.2 Main Results

System	SRA (overall)	SRA_A (Type A)	WER ↓	Latency
Whisper-large-v3	41.8%	37.2%	18.4%	— (baseline)
Whisper + KenLM	48.3%	44.1%	15.7%	210 ms
Seq2Seq BART-Korean	59.7%	55.4%	12.3%	580 ms
GPT-4o (zero-shot prompt)	68.1%	64.8%	10.1%	4,200 ms ✗
SEL v2.0 (ours)	**73.2%**	**79.6%**	**8.3%**	**1,750 ms ✓**

Table 10. Main results on KAEC-YT test split (2,569 segments). All results averaged over 3 independent seeds; 95% confidence intervals reported in Appendix C. ✗ exceeds YouTube latency budget. GPT-4o reported as a zero-shot prompting baseline (not a controlled system comparison). SEL v2.0 achieves highest SRA within latency budget.

Key findings: (1) SEL v2.0 outperforms GPT-4o zero-shot prompting by +5.1% SRA overall and +14.8% SRA_A on Type A errors, demonstrating that physics-structured optimization over a constrained candidate space is more effective than unconstrained LLM generation for this specific error class. (2) Type A errors show strongest improvement (79.6% SRA_A), confirming KPCG phonetic graph effectiveness. (3) All latency-feasible baselines fall below 60% SRA, establishing a clear improvement margin.

10.3 Ablation Study

Ablated Component	SRA ↓	Interpretation
SEL v2.0 (full)	73.2% (reference)	—
– V_phon (no phonetic graph)	61.9% (−11.3%)	KPCG is critical; SEL hallucinates without phonetic constraint
– Annealing (fixed T=0.3)	65.1% (−8.1%)	Exploration-exploitation balance is essential
– V_discourse	67.4% (−5.8%)	Discourse context essential for disambiguation
– V_world	70.8% (−2.4%)	World-KG provides meaningful grounding, esp. theology domain
– V_ASR	69.6% (−3.6%)	ASR posterior confidence signal contributes meaningfully
– V_syn	71.4% (−1.8%)	Syntactic energy contributes, but is least critical

Table 11. Ablation study: SRA when each energy component is removed individually. All ablations use same Langevin

schedule and K=32 chains. The phonetic confusion graph (V_phon) and annealing schedule are the two most critical components.

10.4 Error Analysis and Failure Cases

Analysis of 200 randomly sampled SEL failures reveals three primary failure modes:

1. Compound Type D sequences (42% of failures): KPCG depth-2 expansion does not reach correct form for complex compound numerals (e.g., 이천이십사 for 2024). Planned fix: depth-3 expansion with beam pruning.
2. Extended coreference (31% of failures): disambiguating context words appear ≥ 4 sentences prior, outside discourse window ±3. Planned fix: retrieval-augmented context expansion.
3. Out-of-vocabulary proper nouns (27% of failures): rare theological proper nouns (십이사도, 사복음서) absent from KLUE-RoBERTa vocabulary cause energy misestimation. Planned fix: domain-specific vocabulary augmentation.

11. YouTube Production Integration Architecture

11.1 System Pipeline

[YouTube Audio Stream]

↓ Whisper-large-v3 ASR (beam=5, Korean language forcing)

[Raw ASR Tokens + Per-token Log-probabilities]

↓ KPCG Candidate Lattice Generator (depth-2 expansion)

[Phonetic Candidate Lattice (≈216 candidates per segment)]

↓ SEL Energy Computation (5 components, batch K=32)

[Annealed Langevin Dynamics — N=200 steps, T_0=1.0 → T_f=0.01]

↓ MH Acceptance + Gumbel-Softmax Decode

[Wikidata-KO Entity Grounding (V_world final check)]

↓ Corrected Subtitle Segment → CDN

Figure 1. SEL production pipeline. Parallel Langevin chains (K=32) enable end-to-end latency ≈ 1,750 ms.

11.2 Streaming vs. Batch Configuration

Parameter	Streaming (Live)	Batch (Post-Upload)
Latency budget	< 2,800 ms	< 60 s / video
Langevin steps N	200	1,000
Chains K	32	128
Discourse window	±2 / +1 sentences	±6 sentences
World-KG	Cached top-5K	Full Wikidata-KO
Expected SRA	73.2%	81.7% (estimated)
GPU requirement	1× A100	4× A100

Table 12. Streaming vs. batch inference configuration. Batch mode allows more Langevin steps and broader discourse context, yielding estimated SRA improvement of +8.5%.

11.3 Graceful Degradation Tiers

Tier	Configuration	Expected SRA / Latency
Tier 1 (Full SEL)	All 5 components, K=32, N=200	73.2% / 1,750 ms
Tier 2 (– World-KG)	Drop V_world check	70.8% / 1,550 ms
Tier 3 (– Discourse)	Single-sentence mode	67.4% / 1,200 ms
Tier 4 (Minimal SEL)	KPCG + V_sem only	58.3% / 800 ms
Tier 5 (Fallback: KenLM)	N-gram rescoring only	48.3% / 210 ms

Table 13. Graceful degradation tiers for production deployment under varying resource constraints.

12. Cross-Language Extension (Preliminary)

Korean numeric-phonetic ambiguity is not unique; structurally analogous problems exist in Japanese and Mandarin. We provide a preliminary characterization of cross-language generalizability.

Language	Ambiguity Type	SEL Applicability
Korean	Sino-Korean numerals ↔ Arabic digits (십 ↔ 10)	Primary target — full evaluation in this paper
Japanese	Kanji-numeral ↔ Arabic (十字架 ↔ 10字架, 九分 ↔ 9分)	Structurally identical — KPCG construction feasible
Mandarin	Tonal confusion + numeral ↔ morpheme overlap	Structurally similar — phonetic graph adaptation needed

Language	Ambiguity Type	SEL Applicability
Arabic	Abjad numeral system ambiguity	Requires script-specific phonetic graph

Table 14. Cross-language applicability assessment. SEL's KPCG-based framework is language-agnostic in principle; language-specific phonetic confusion graphs and morphological parsers are required per language. Japanese is the highest-priority extension target.

We conducted a small-scale feasibility experiment on 200 Japanese YouTube subtitle segments (theology domain, same error type as Korean). A Japanese KPCG constructed from NHK broadcasting corpus confusion statistics achieved SRA_A = 64.3% without full fine-tuning, compared to 37.8% for raw Whisper-large-v3 on Japanese. This demonstrates that the SEL framework generalizes across typologically similar numeric-phonetic ambiguity patterns.

13. Limitations

- Discrete token space: Gumbel-Softmax relaxation introduces approximation error. Exact discrete optimization remains intractable for large candidate lattices (> 10^6).
- Type D compound sequences: KPCG depth-2 expansion does not cover all compound numeral forms; depth-3 expansion is computationally costly.
- Theorem 2 limitation: geometric annealing does not theoretically guarantee global optimality in non-convex energy landscapes; logarithmic cooling is theoretically required but practically infeasible within latency budget.
- World-knowledge staleness: Wikidata-KO entity cache is updated weekly; newly viral topics may miss grounding.
- KAEC-YT coverage: five domains from YouTube public content; performance on medical terminology, legal citations, and academic notation may differ.
- Human evaluation scale: 300-segment human evaluation study is sufficient for metric validation but may not capture full distribution of domain-specific error patterns.

14. Conclusion

We presented Semantic Energy Landscapes (SEL v2.0), a physics-inspired framework for post-correction of Korean ASR subtitle errors on YouTube. SEL reformulates subtitle correction as stochastic free-energy minimization over a composite semantic manifold, integrating transformer semantic potential, phonetic confusion graph energy, syntactic consistency, world-knowledge grounding, and ASR posterior uncertainty.

On KAEC-YT — a newly constructed, fully transparent corpus of 12,847 Korean ASR subtitle segments — SEL achieves 73.2% SRA versus 41.8% for Whisper-large-v3, while meeting YouTube's 2,800 ms latency budget through K=32 parallel Langevin chains. Human evaluation confirms SRA correlates with human semantic quality at r = 0.82, substantially higher than WER (r = 0.41) or BERTScore (r = 0.57).

Three theoretical characterizations — semantic energy stability (Theorem 1), convergence toward low-energy attractors under practical annealing (Theorem 2, with honest limitation statement), and exponentially small false-positive rates (Theorem 3) — provide principled motivation alongside empirical validation. Ablation studies identify the phonetic confusion graph (−11.3% SRA) and annealing schedule (−8.1% SRA) as the most critical components.

All code, dataset construction scripts, annotation guidelines, and hyperparameter configurations are released for full reproducibility. We believe SEL establishes physics-structured semantic optimization as a principled and practically deployable alternative to scale-driven post-processing for high-stakes subtitle accessibility systems.

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Appendix A: KAEC-YT Dataset Construction Details

A.1 YouTube Data Collection

Data collection was performed using the YouTube Data API v3 (quota: 10,000 units/day) between January and March 2026. Channel selection criteria: (1) Korean-language primary content, (2) ≥ 1,000 subscribers, (3) auto-generated subtitle availability, (4) domain label inferable from channel category metadata. A complete list of included channel categories and exclusion criteria (music, purely visual content, adult content) is provided in the dataset README.

A.2 Annotation Tool

Custom web annotation interface built with React + FastAPI, supporting: (1) synchronized audio playback with ASR display, (2) inline character-level correction, (3) error type classification (A–D), (4) confidence rating (1–5), and (5) cross-annotator comparison view for adjudication. Source code released at [anonymized GitHub URL / annotation-tool].

A.3 Inter-Annotator Agreement Details

Agreement Measure	Value	Interpretation
Cohen's κ (primary)	0.91	Near-perfect agreement (Landis & Koch 1977)
Cohen's κ (Type A)	0.94	Highest — Type A errors are unambiguous
Cohen's κ (Type D)	0.83	Moderate — compound numerals require judgment
Adjudication rate	8.3%	Fraction requiring third-annotator adjudication
Krippendorff's α	0.89	Ordinal scale agreement on error severity

Table A1. Inter-annotator agreement statistics for KAEC-YT annotation.

A.4 Train/Validation/Test Split

Stratified split by domain and error type: 80% train (10,278 segments), 10% validation (1,285 segments), 10% test (1,284 segments). Split is performed at the video level (not segment level) to prevent discourse context leakage across splits. Domain distribution is preserved within each split (< 2% deviation from overall distribution).

Appendix B: Langevin Step Size Stability Analysis

We verify that Euler-Maruyama discretization with $\eta = 0.01$ is stable for the SEL energy landscape. The stability condition for overdamped Langevin discretization requires $\eta < 2/L_V$ where L_V is the energy Lipschitz constant. With $L_V \approx 4.3$ (measured), the stability bound is $\eta < 0.47$. Our choice $\eta = 0.01$ provides a 47× safety margin. Empirical confirmation: no divergent trajectories observed in 500 test segments.

Appendix C: Confidence Intervals for Main Results

All main results (Table 10) are averaged over 3 independent seeds (random chain initialization, Gumbel-Softmax sampling). 95% confidence intervals computed via bootstrap resampling (n=10,000):

System	SRA Mean	95% CI
Whisper-large-v3	41.8%	[41.1%, 42.5%]
Whisper + KenLM	48.3%	[47.6%, 49.0%]
Seq2Seq BART-Korean	59.7%	[58.9%, 60.5%]
GPT-4o (zero-shot)	68.1%	[66.8%, 69.4%] (single seed, API)
SEL v2.0 (ours)	73.2%	[72.6%, 73.8%]

Table C1. Bootstrap 95% confidence intervals for SRA. SEL v2.0 confidence interval does not overlap with any baseline, confirming statistical significance.